Lecture 3: Bargaining games

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Lecture 3 Outline

- Infinite horizon Alternating-offer Bargaining model
- Breakdown points, outside and inside options
- Renegotiable model for Shapley value
- Non-renegotiable model for Nucleolus value
- Extra Reading:
  - Muthoo (1999). Bargaining theory with applications.
  - Stole and Zwiebel (1996). Intra-firm bargaining under non-binding contracts. The Review of Economic Studies.
  - Montez (2014). One-to-many bargaining when pairwise agreements are non-renegotiable. Journal of Economic Theory.



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Negotiation

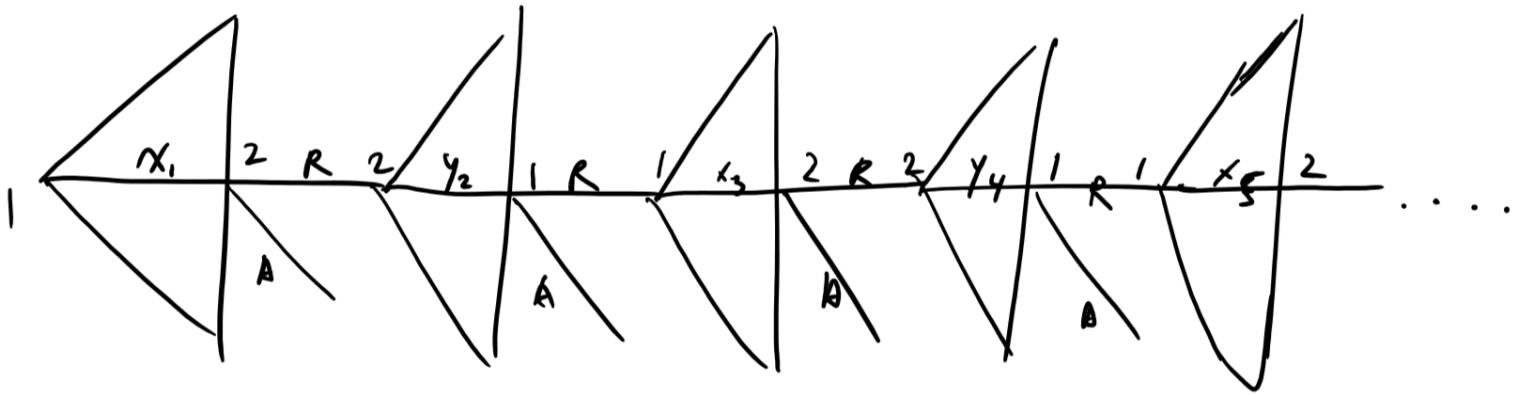
- Use the advantage/disadvantage (Video)
- Create the advantage/disadvantage (Video)



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Infinite horizon Alternating-offer Bargaining model

- $T = \infty$
- Players alternate in proposing and responding
- Every period is discounted

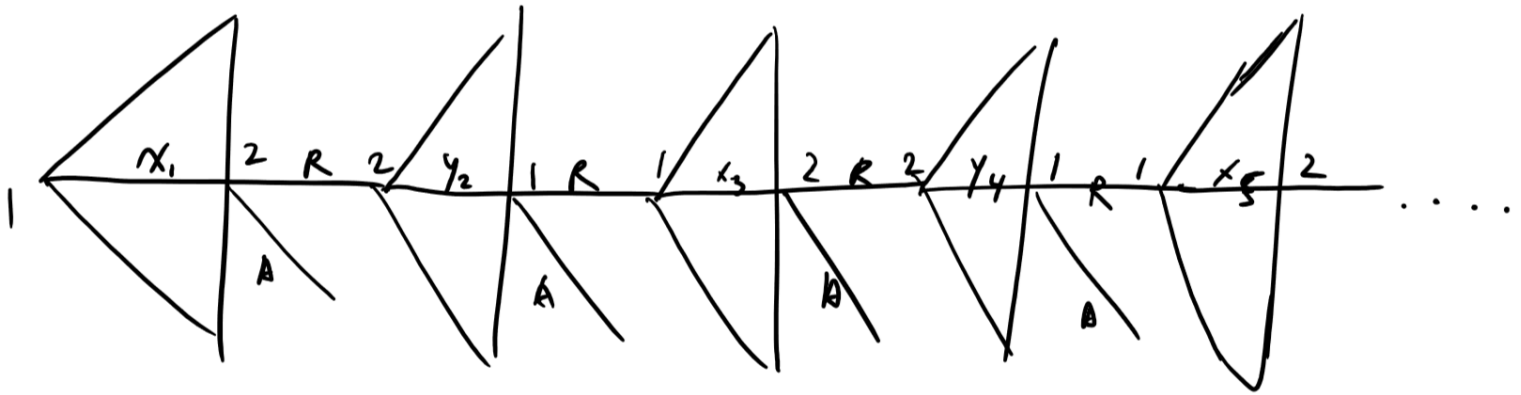


- The great Rubinstein (1982)
- https://arielrubinstein.tau.ac.il/

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Infinite horizon Alternating-offer Bargaining model

- No agreement is not a Nash equilibrium.
- Each player is in the identical situation when he (she) is proposing or responding: past history does not matter, future looks exactly the same.
- Therefore, in equilibrium, each player makes the a minimum acceptable offer x^* and y^*, respectively, whenever they get to propose.



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Infinite horizon Alternating-offer Bargaining model

- Therefore, in each stage of the game, each player is indifferent between accepting the received offer and rejecting the offer to make his/her own in the next stage:

$$\begin{cases} x^* = \delta_2(1 - y^*) \\ y^* = \delta_1(1 - x^*) \end{cases}$$

- In equilibrium:

$$x^* = \frac{\delta_2(1 - \delta_1)}{1 - \delta_1\delta_2}$$

$$y^* = \frac{\delta_1(1 - \delta_2)}{1 - \delta_1\delta_2}$$

- How to correctly interpret x^* and y^*?

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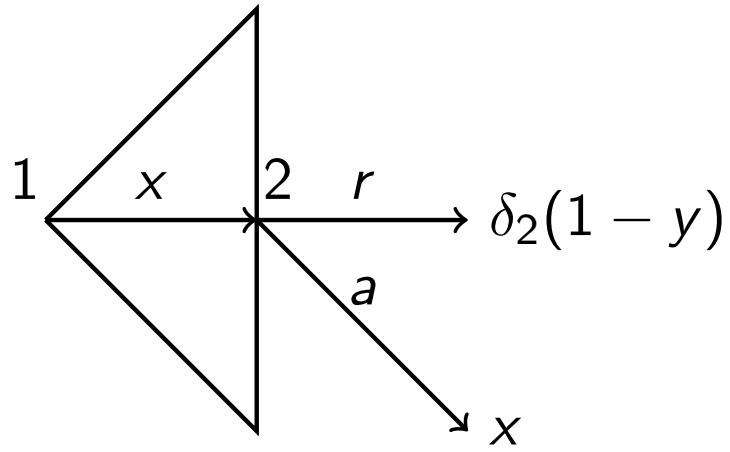
Infinite horizon Alternating-offer Bargaining model

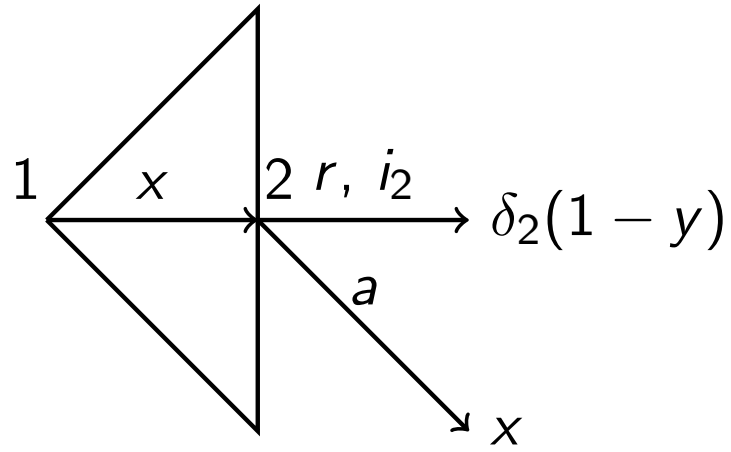
- Looks like the bargaining power of each player merely depends on how patient they are. That seems to be an over-simplification.
- So let us think of something else that can possibly affect the result:
  - Breakdown points: the payments parties have if the negotiation gets interrupted and they are unable to reach an agreement yet.
  - Inside option: the temporary payments parties earn during their negotiation before reaching an agreement.
  - Outside option: the alternative payments parties can gain if they proactively end the negotiation in advance.

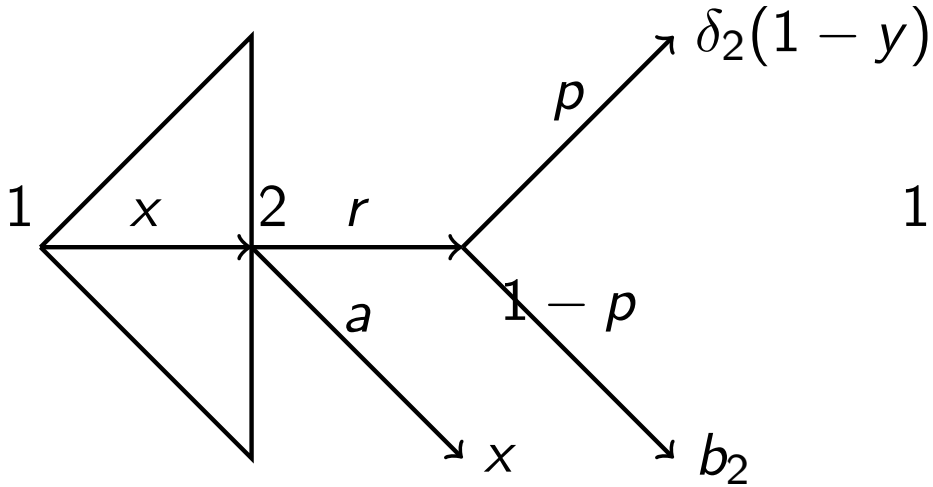


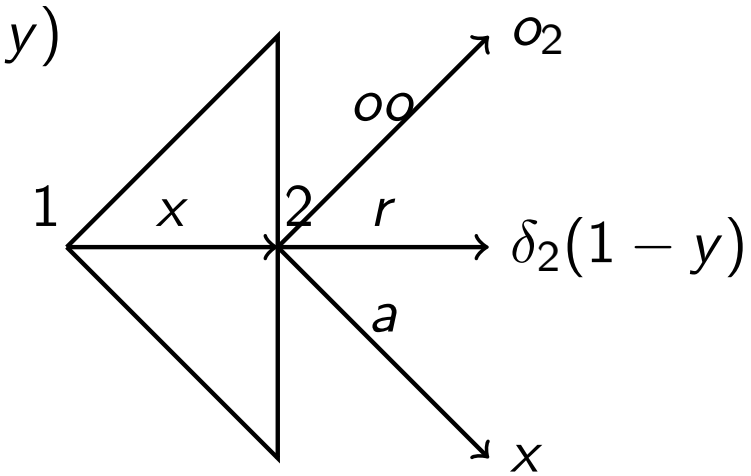
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Breakdown points, inside option and outside option









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Breakdown points

- In Rubinstein's structure, assume that after each period, the game continues with a probability p and breaks down with a probability $1-p$, leading to two breakdown points b_1 and b_2, $b_1 + b_2 < 1$.
- Indifferent functions become:

$$\begin{cases} x^* = \delta_2(p(1-y^*) + (1-p)b_2) \\ y^* = \delta_1(p(1-x^*) + (1-p)b_1) \end{cases}$$

- For simplicity, suppose that $\delta_1 = \delta_2 = 1$, in which case the equilibrium outcome becomes:

$$x^* = \frac{p(1-b_1) + b_2}{1+p}$$

$$y^* = \frac{b_1 + p(1-b_2)}{1+p}$$



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Inside option

- The payoffs that the players obtain when the negotiation continues are their inside options i_1 and i_2, respectively. Consider this in Rubinstein's structure, indifferent functions become:

$$\begin{cases} x^* = \delta_2(1 - y^*) + i_2 \\ y^* = \delta_1(1 - x^*) + i_1 \end{cases}$$

- The equilibrium outcome becomes:

$$x^* = \frac{\delta_2(1 - \delta_1)}{1 - \delta_1\delta_2} + \frac{i_2 - \delta_2 i_1}{1 - \delta_1\delta_2}$$

$$y^* = \frac{\delta_1(1 - \delta_2)}{1 - \delta_1\delta_2} + \frac{i_1 - \delta_1 i_2}{1 - \delta_1\delta_2}$$



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Outside option

- The payoffs that the players obtain while they choose to terminate the negotiation are their outside options o_1 and o_2, respectively. Consider this in Rubinstein's structure, indifferent functions become:

$$\begin{cases} x^* = max(\delta_2(1 - y^*), o_2) \\ y^* = max(\delta_1(1 - x^*), o_1) \end{cases}$$

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Outside option

- Take $\Delta_1 = \frac{1-\delta_2}{1-\delta_1\delta_2}$ and $\Delta_2 = \frac{1-\delta_1}{1-\delta_1\delta_2}$
- The equilibrium outcome becomes:

$$x^* = \begin{cases} \delta_2\Delta_2 & \text{if } o_1 \leq \delta_1\Delta_1 \text{ and } o_2 \leq \delta_2\Delta_2 \\ o_2 & \text{if } o_1 \leq \delta_1(1-o_2) \text{ and } o_2 > \delta_2\Delta_2 \\ \delta_2(1-o_1) & \text{if } o_1 > \delta_1\Delta_1 \text{ and } o_2 \leq \delta_2(1-o_1) \\ o_2 & \text{if } o_1 > \delta_1(1-o_2) \text{ and } o_2 > \delta_2(1-o_1) \end{cases}$$

$$y^* = \begin{cases} \delta_1\Delta_1 & \text{if } o_1 \leq \delta_1\Delta_1 \text{ and } o_2 \leq \delta_2\Delta_2 \\ \delta_1(1-o_2) & \text{if } o_1 \leq \delta_1(1-o_2) \text{ and } o_2 > \delta_2\Delta_2 \\ o_1 & \text{if } o_1 > \delta_1\Delta_1 \text{ and } o_2 \leq \delta_2(1-o_1) \\ o_1 & \text{if } o_1 > \delta_1(1-o_2) \text{ and } o_2 > \delta_2(1-o_1) \end{cases}$$



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Outside option

- Suppose that $\begin{cases} \delta_2(1-y^*) > o_2 \\ \delta_1(1-x^*) > o_1 \end{cases}$, indifferent function becomes $\begin{cases} x^* = \delta_2(1-y^*) \\ y^* = \delta_1(1-x^*) \end{cases}$ and the equilibrium becomes $\begin{cases} x^* = \frac{\delta_2(1-\delta_1)}{1-\delta_1\delta_2} = \delta_2\Delta_2 \\ y^* = \frac{\delta_1(1-\delta_2)}{1-\delta_1\delta_2} = \delta_1\Delta_1 \end{cases}$. Substitute into the supposition, we can solve and see that $\begin{cases} o_2 < \delta_2(1-\frac{\delta_1(1-\delta_2)}{1-\delta_1\delta_2}) = \delta_2\Delta_2 \\ o_1 < \delta_1(1-\frac{\delta_2(1-\delta_1)}{1-\delta_1\delta_2}) = \delta_1\Delta_1 \end{cases}$.
- The intuition is that, when the outside options are both relatively small, players would never opt for the outside options and the negotiation proceeds as if the outside options do not exist.



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Outside option

- Suppose that $\begin{cases} \delta_2(1-y^*) < o_2 \\ \delta_1(1-x^*) > o_1 \end{cases}$, indifferent function becomes $\begin{cases} x^* = o_2 \\ y^* = \delta_1(1-x^*) \end{cases}$ and the equilibrium becomes $\begin{cases} x^* = o_2 \\ y^* = \delta_1(1-o_2) \end{cases}$. Substitute into the supposition, we have $\begin{cases} \delta_2(1-\delta_1(1-o_2)) < o_2 \\ \delta_1(1-o_2) > o_1 \end{cases}$, or $\begin{cases} o_1 < \delta_1(1-o_2) \\ o_2 > \delta_2\Delta_2 \end{cases}$.
- In this case, only player 2's outside option is large enough to be effective.

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Outside option

- Suppose that $\begin{cases}\delta_2(1-y^*) > o_2 \\ \delta_1(1-x^*) < o_1\end{cases}$, indifferent function becomes $\begin{cases}x^* = \delta_2(1-y^*) \\ y^* = o_1\end{cases}$ and the equilibrium becomes $\begin{cases}x^* = \delta_2(1-o_1) \\ y^* = o_1\end{cases}$. Substitute into the supposition, we have $\begin{cases}\delta_2(1-o_1) > o_2 \\ \delta_1(1-\delta_2(1-o_1)) < o_1\end{cases}$, or $\begin{cases}o_1 > \delta_1\Delta_1 \\ o_2 < \delta_2(1-o_1)\end{cases}$.
- In this case, only player 1's outside option is large enough to be effective.

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Outside option

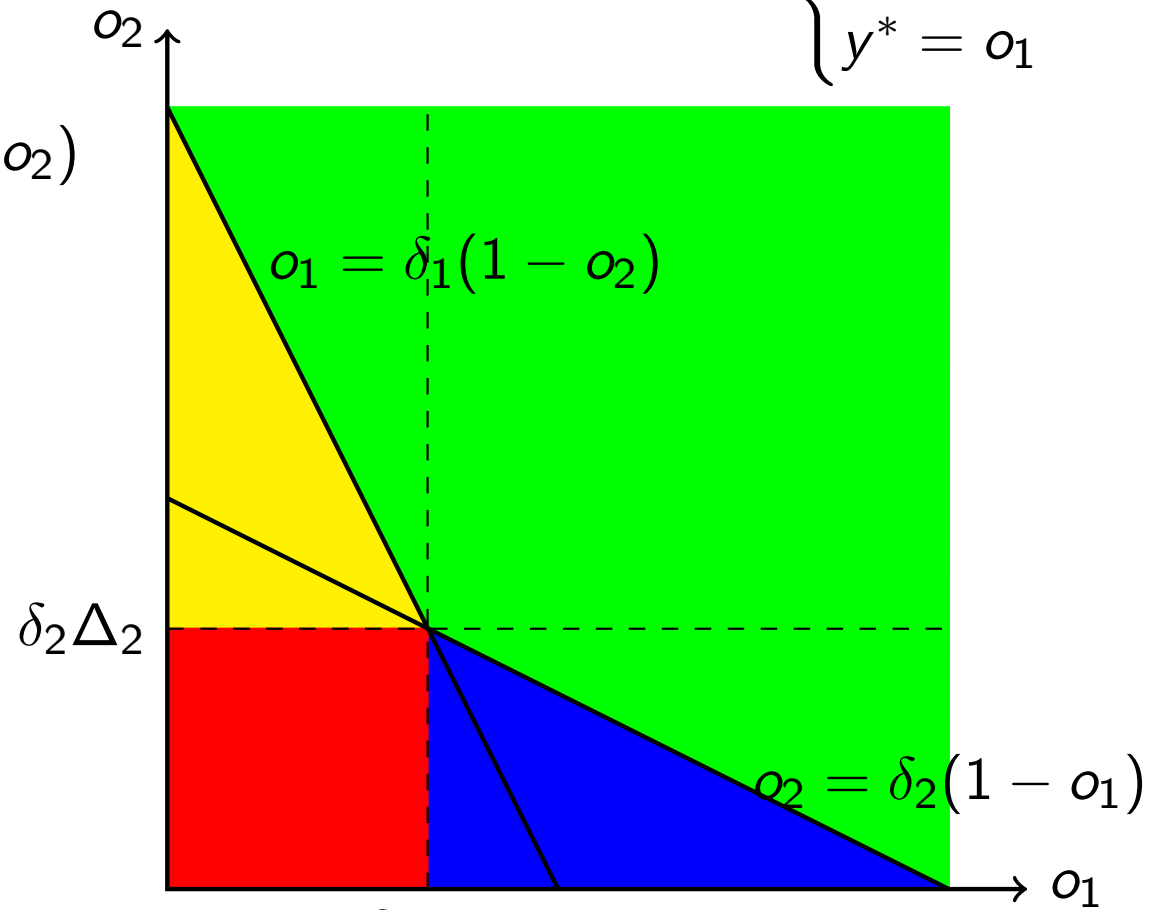
- Suppose that $\begin{cases} \delta_2(1-y^*) < o_2 \\ \delta_1(1-x^*) < o_1 \end{cases}$, indifferent function becomes $\begin{cases} x^* = o_2 \\ y^* = o_1 \end{cases}$ and the equilibrium is the same.
  Substitute into the supposition, we can solve and see that $\begin{cases} o_2 > \delta_2(1-o_1) \\ o_1 > \delta_1(1-o_2) \end{cases}$.
- In this case, when both outside options are large enough, players will opt out with no hesitation at all.

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Outside option

$$\begin{cases} x^* = o_2 \\ y^* = \delta_1(1 - o_2) \end{cases}$$

$$\begin{cases} x^* = o_2 \\ y^* = o_1 \end{cases}$$



$$\begin{cases} x^* = \delta_2\Delta_2 \\ y^* = \delta_1\Delta_1 \end{cases}$$

$$\begin{cases} x^* = \delta_2(1 - o_1) \\ y^* = o_1 \end{cases}$$

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A summary

- Player i's equilibrium outcome increases with d_i and decrease with d_j.
- Player i's equilibrium outcome increases with i_i and decrease with i_j.
- Player i's equilibrium outcome increases with o_i and decrease with o_j.
- For example. In a university strike: workers (i) can beat the university (j) if there is a limited provision of alternatives in this year's fresh PHD market and they can easily find better jobs (high d_i) but the university cannot find better teachers (low d_j), if they have payment elsewhere (high i_i) but the university keeps refunding tuition (low i_j), if they already hold offers (high o_i) but the university has not found any replacement yet (low o_j).



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Multiple-players negotiation

- Until now, we are limiting our discussion between two-players negotiations. However, things could easily get more complicated.
- Multi-lateral negotiation: 1+4+6 wechat groups in a 4-people dormitory room
- Multi-lateral negotiation: unanimous rule and majority rule
- In this chapter, we discuss one negotiation between more than two players: one principle and more than one (two) agents.

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One principal two agents example

- To make the multiple-players negotiations possible, we construct multiple-period negotiation protocols. In each period, P negotiates with one of the agent: N-players to two-players
- We provide two different protocols, one is called Renegotiable protocol and the other called Non-renegotiable protocol.
- Think of a big round table with N players. Renegotiable protocol: leave the table when negotiation fails; non-renegotiable protocol: leave the table when negotiation succeeds.
- Potential issues: irrevocability of the offers, over-promising...

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One principal two agents example

- Recall the assumption: P could hire A_1 to create a value of $V_{P1} = a$, hire A_2 to create a value of $V_{P2} = b$ or hire both to create a value of $V_{P12} = 1$. No one can work alone and A_1 and A_2 can not work without P involved.
- Let d_i denotes player i's disagreement payoff after each period's negotiation. Let x and y denote the value P offers to A_1 and A_2, respectively.



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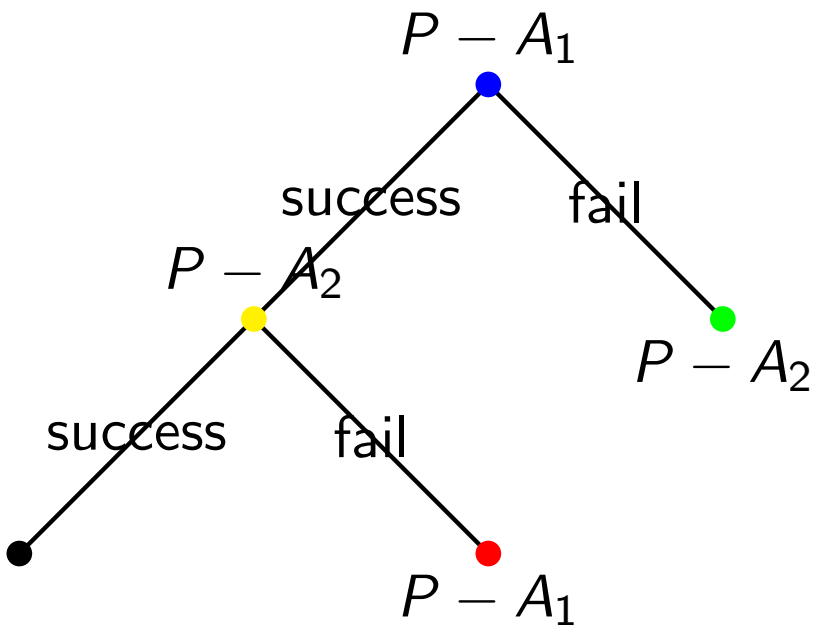
One principal two agents example

- Split-the-difference-rule: for convenience, assume that in each round of negotiation between two players, they split the distributable pie equally, meaning that for each player, their gaining from the agreement minus the disagreement payoff is the same. In other words, the agreement equally benefits the two players.
- For example, if two players can create a value of 1 together when an agreement is reached, and each gets 0.2 and 0.6 when negotiation fails, respectively, then the agreement outcome will be 0.3 and 0.7, as $0.3 - 0.2 = 0.7 - 0.6$.

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Renegotiable protocol

- P negotiates with agents in sequence. When negotiation succeeds, a temporary agreement is reached. When negotiation fails, the agent leaves the table, while revoking all the temporal agreements signed before.
- In this protocol, re-negotiation is possible. The game will go on at most three periods.



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Sub-game perfect Nash equilibrium

- Suppose the negotiation succeeds in node blue and P promises to pay x to A_1. If negotiation fails in node yellow, then A_2 exits, x is overturned. In node red, the pie shrinks to a, the negotiation will succeed: P pays $\frac{a}{2}$ to A_1.
- Anticipating this, in node yellow, $d_p = \frac{a}{2}$ and $d_2 = 0$. Between P and A_2: $1 - x - y - \frac{a}{2} = y - 0$, P pays $y = \frac{1-x-\frac{a}{2}}{2}$ to A_2.
- If negotiation fails in node blue, then A_1 exits, in node green the pie shrinks to b, the negotiation will succeed: P pays $\frac{b}{2}$ to A_2.



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Sub-game perfect Nash equilibrium

- Anticipating all the future, in node blue, P's agreement payoff is $1-x-y=\frac{1-x+\frac{a}{2}}{2}$ and $d_P=\frac{b}{2}$, while 1's agreement payoff is x and $d_1=0$.
- Therefore, in node blue, between P and A_1: $\frac{1-x+\frac{a}{2}}{2}-\frac{b}{2}=x-0$, it can be easily derived that $x=\frac{2+a-2b}{6}$.
- With the value of x, we can further calculate that $y=\frac{1-x-\frac{a}{2}}{2}=\frac{2+b-2a}{6}$. The final agreement is $(1-x-y,x,y)=(\frac{2+a+b}{6},\frac{2+a-2b}{6},\frac{2+b-2a}{6})$.

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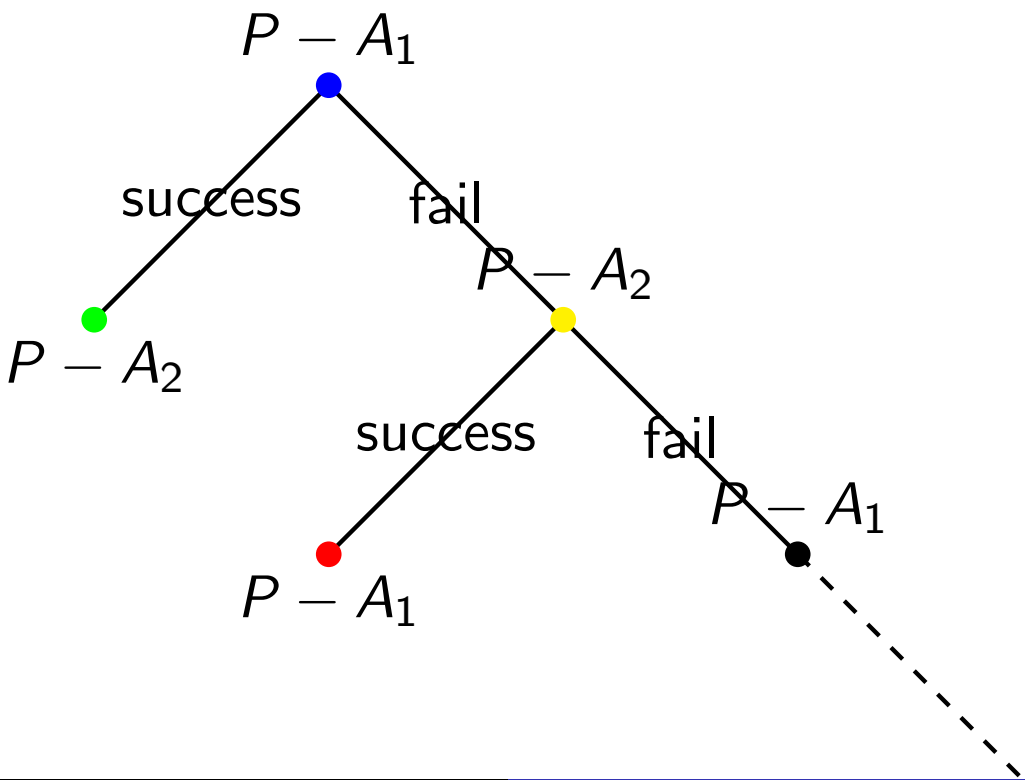
Shapley Value

- When $a = \frac{1}{2}$ and $b = \frac{2}{3}$,
  $(\frac{2+a+b}{6}, \frac{2+a-2b}{6}, \frac{2+b-2a}{6}) = (\frac{19}{36}, \frac{7}{36}, \frac{10}{36})$.
- Therefore, from the renegotiable protocol, we can "induce" negotiators to reach the Shapley value distribution.

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Non-renegotiable protocol

- P negotiates with agents in sequence. When negotiation succeeds, the deal is sealed and the agent leaves the table. When negotiation fails, the agent goes to the end of the queue.
- In this protocol, re-negotiation is impossible. The game could go on forever.



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Sub-game perfect Nash equilibrium

- If the negotiation succeeds in node blue and P promises to pay x to A_1, the game proceeds to node green, where $d_p = max(a - x, 0)$ and $d_2 = 0$: if A_2 doesn't sign, P can fulfill his commitment to A_1 if $x < a$, but he can't if $x > a$. P gets a payoff of $a - x$ when he can and 0 otherwise.
- Therefore, in node green, between P and A_2: $1 - x - y - max(a - x, 0) = y - 0$, P pays $y = \frac{1-x-max(a-x,0)}{2}$ to A_2. The agreement in node green will be $(\frac{1-x+max(a-x,0)}{2}, x, \frac{1-x-max(a-x,0)}{2})$.

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Sub-game perfect Nash equilibrium

- Similarly, if the negotiation fails in node blue, A_1 goes to the end of the queue (node black), the game proceeds to node yellow. Suppose P promises to pay y to A_2, then $d_p = max(b-y, 0)$ and $d_1 = 0$ in node red, where P pays $x = \frac{1-y-max(b-y,0)}{2}$ to A_1. The agreement in node red will be $(\frac{1-y+max(b-y,0)}{2}, \frac{1-y-max(b-y,0)}{2}, y)$.
- In node blue, players are aware of the situation: the game will proceed to node green if negotiation succeeds and to node red (through node yellow) if negotiation fails. Therefore, in node blue, between P and A_1, we have the equal split: $\frac{1-x+max(a-x,0)}{2} - \frac{1-y+max(b-y,0)}{2} = x - \frac{1-y-max(b-y,0)}{2}$, viz., $1 = 3x + 2max(b-y, 0) - max(a-x, 0)$.

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Nucleolus Value

- Similarly, at node yellow, between P and A_2:
  $1 = 3y + 2max(a - x, 0) - max(b - y, 0)$.
- Notice that the game is infinite and symmetric. Combine the two equations, we have $\begin{cases} x = \frac{1-y-max(b-y,0)}{2} \\ y = \frac{1-x-max(a-x,0)}{2} \end{cases}$.
- It can be easily verified that the final agreement is $(\frac{7}{12}, \frac{1}{6}, \frac{1}{4})$ when $a = \frac{1}{2}$ and $b = \frac{2}{3}$.

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Notice

- The two cooperative values (Shapley and Nucleolus) are widely applicable concepts.
- The two non-cooperative approach we introduce here, instead, are only applicable in the one principal, two (multiple) agents model—implementation is always more challenging.



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Thank you



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